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# **On the Probability Generating Function of Discrete Distribution Generated by Cauchy Stable Density**

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## **Abstract:**

In this paper, we study the probability generating function (PGF) of a discrete distribution induced by the Cauchy stable density, referred to as the Discrete Cauchy Distribution. An approximate analytical representation of the PGF is derived, and its asymptotic behavior is examined. The analysis shows that the PGF develops a logarithmictype singularity in its domain, indicating that the distribution is heavy-tailed with infinite mean. These results provide a precise analytical characterization of the Discrete Cauchy Distribution and offer useful insights for modeling and inference in settings involving heavy-tailed count data.

**Keywords:** Discrete Cauchy distribution, Probability generating function, Asymptotic analysis.

**Mathematics Subject Classification (2010):** 60E05, 60E10, 60E07, 62E10.

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## **1 Introduction**

Discrete heavy-tailed distributions arise naturally in many areas of probability theory and applied statistics. Such models are particularly suitable for describing phenomena where extreme values occur with non-negligible probability, a feature that frequently appears in complex biological systems, network dy-

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namics, and financial data. In these contexts, classical light-tailed discrete models often fail to adequately capture the variability observed in empirical data [Foss et al. \(2013\)](#); [Kotz et al. \(2005\)](#); [Kuznetsov \(2017\)](#); [Kuznetsov et al. \(2022\)](#).

A one-parameter discrete distribution generated by Cauchy stable density (*Discrete Cauchy Distribution*), was introduced in [Farbod and Gasparian \(2008\)](#). This model was motivated by irregular patterns observed in bioinformatics and biological frequency data. Subsequent studies investigated statistical inference and asymptotic properties of this distribution, including maximum likelihood estimation and related inferential procedures [Farbod and Gasparian \(2008\)](#); [Farbod et al. \(2010\)](#). Due to its heavy-tailed nature, the model shares several qualitative properties with the continuous Cauchy distribution, such as slow decay of probabilities and the absence of finite moments.

Despite these developments, several analytical aspects of the distribution remain insufficiently explored. In particular, functional representations of the model have not been extensively studied. One of the most useful analytical tools for investigating discrete random variables is the probability generating function (PGF). For a non-negative integer-valued random variable  $X$  with probability mass function (PMF)  $p_x$ , the PGF is defined as

$$G_X(t) = \mathbb{E}(t^X) = \sum_{x=1}^{\infty} p_x t^x, \quad |t| \leq 1.$$

The generating function provides a convenient framework for analyzing structural properties of discrete distributions. In particular, the behavior of the PGF near  $t = 1$  carries important information about moment existence and tail characteristics. Singular behavior of generating functions often reflects heavy-tailed properties of the underlying distribution.

The main objective of this paper is to study the PGF of the Discrete Cauchy distribution. We derive an approximate analytical representation of the PGF and investigate its asymptotic behavior in a neighborhood of  $t = 1$ . The obtained results reveal how the heavy-tailed structure of the PMF manifests itself in the singular structure of the generating function. These findings provide additional insight into the probabilistic structure of discrete distribution generated by Cauchy stable density.

## 2 The PGF

Consider the following Cauchy stable density [Astola and Danielian \(2007\)](#); [Nolan \(2020\)](#); [Zolotarev \(1986\)](#)

$$f_x(\gamma) = \frac{\gamma}{2 \left( \frac{\pi^2}{4} \gamma^2 + x^2 \right)}, \quad x > 0, \quad \gamma > 0. \quad (2.1)$$

with location parameter equal to zero, where  $\gamma$  denotes the scale parameter.

Using (2.1), the PMF of the Discrete Cauchy Distribution is constructed as follows [Farbod and Gasparian \(2008\)](#)

$$p_x(\gamma) = \frac{1}{c(\gamma)} f_x(\gamma), \quad x = 1, 2, \dots, \quad \gamma > 0, \quad (2.2)$$

where  $c(\gamma)$  is the normalization constant given by

$$c(\gamma) = \sum_{y=1}^{\infty} f_y(\gamma) = \sum_{y=1}^{\infty} \frac{\gamma}{2 \left( \frac{\pi^2}{4} \gamma^2 + y^2 \right)}. \quad (2.3)$$

We refer to the one-parameter distribution defined in (2.2) as the *Discrete Cauchy Distribution*. For large values of  $x$ , the tail of the distribution satisfies

$$p_x(\gamma) \sim \frac{K}{x^2},$$

where the notation  $\sim$  denotes asymptotic equivalence as  $x \rightarrow \infty$ , and  $K$  is a positive constant depending on  $\gamma$ . This behavior indicates that the model defined in (2.2) possesses a heavy tail. Motivated by this property, we investigate the PGF of the distribution.

The PGF of Discrete Cauchy Distribution (2.2) is defined by

$$G_X(t) = \mathbb{E}(t^X) = \sum_{x=1}^{\infty} p_x(\gamma) t^x, \quad |t| \leq 1. \quad (2.4)$$

Let us first establish the following lemma.

**Lemma 2.1.** *For every  $0 \leq t < 1$ , the PGF  $G_X(t)$  is well defined and the series in (2.4) converges absolutely.*

**Proof.** Since  $p_x(\gamma) \sim \frac{K}{x^2}$  as  $x \rightarrow \infty$ , there exists a constant  $C > 0$  such that  $|p_x(\gamma)| \leq \frac{C}{x^2}$ , for sufficiently large  $x$ . Hence

$$|p_x(\gamma) t^x| \leq \frac{C t^x}{x^2}.$$

Because the series  $\sum_{x=1}^{\infty} \frac{t^x}{x^2}$  converges for  $0 \leq t < 1$ , it follows by comparison that the PGF series converges absolutely.  $\square$

**Proposition 2.2.** *The PGF  $G_X(t)$  of the Discrete Cauchy Distribution (2.2) has radius of convergence equal to one.*

**Proof.** Since  $p_x(\gamma) \sim Kx^{-2}p_x(\gamma)$  as  $x \rightarrow \infty$ , we have

$$\limsup_{x \rightarrow \infty} |p_x(\gamma)|^{\frac{1}{x}} = 1.$$

Therefore the power series

$$G_X(t) = \sum_{x=1}^{\infty} p_x(\gamma)t^x$$

has radius of convergence equal to 1. The proof is complete.  $\square$

Substituting  $p_x(\gamma)$  from (2.2) into (2.4), we obtain

$$G_X(t) = \frac{1}{c(\gamma)} \sum_{x=1}^{\infty} \frac{\gamma t^x}{2 \left( \frac{\pi^2}{4} \gamma^2 + x^2 \right)}.$$

For large values of  $x$  we have the approximation

$$\frac{1}{\frac{\pi^2}{4} \gamma^2 + x^2} \sim \frac{1}{x^2}.$$

Hence the tail of the series can be approximated by

$$G_X(t) \approx K(\gamma) \sum_{x=1}^{\infty} \frac{t^x}{x^2},$$

where  $K(\gamma) = \frac{\gamma}{2c(\gamma)}$ . It is well known that

$$\sum_{x=1}^{\infty} \frac{t^x}{x^2} = \text{Li}_2(t),$$

where  $\text{Li}_2(t)$  denotes the dilogarithm (polylogarithm of order 2) [Lewin \(1981\)](#). Therefore an approximate expression for the PGF is

$$G_X(t) \approx K(\gamma) \text{Li}_2(t).$$

## 2.1 Behavior near $t = 1$

A well-known asymptotic expansion of the dilogarithm function near  $t = 1$  is (see, for example, [Lewin \(1981\)](#))

$$\text{Li}_2(t) = \frac{\pi^2}{6} - (1-t) \ln(1-t) + O(1-t), \quad t \rightarrow 1,$$

where  $O(1 - t)$  denotes the Big-O term; that is, there exists a constant  $C > 0$  such that the remainder is bounded in magnitude by  $C(1 - t)$  as  $t \rightarrow 1$ . Consequently, the PGF satisfies

$$G_X(t) \sim K(\gamma) \left[ \frac{\pi^2}{6} - (1 - t) \ln(1 - t) \right], \quad t \rightarrow 1, \quad (2.5)$$

where  $K(\gamma) > 0$  is a constant depending on the scale parameter  $\gamma$ .

The presence of the logarithmic-type term  $(1 - t) \ln(1 - t)$  is consistent with the power-law decay of the PMF and reflects the heavy-tailed nature of the distribution.

**Remark 2.3.** *The term  $(1 - t) \ln(1 - t)$  appearing in the asymptotic expansion of the PGF is characteristic of distributions whose tails behave as  $x^{-2}$ . Such a logarithmic singularity is typical for regularly varying distributions (Bingham et al. (1987)) with tail index two and corresponds to the borderline case between finite and infinite first moments.*

## 2.2 On the mean

Let

$$G'_X(t) = \frac{dG_X(t)}{dt} = \sum_{x=1}^{\infty} x p_x(\gamma) t^{x-1}$$

denote the first derivative of the PGF. Since

$$G'_X(1^-) = \mathbb{E}[X],$$

the existence of the mean is determined by the behavior of  $G_X(t)$  as  $t \rightarrow 1^-$ .

Using the asymptotic expansion (2.5), we obtain

$$G_X(t) \sim K(\gamma) \left[ \frac{\pi^2}{6} - (1 - t) \ln(1 - t) \right], \quad t \rightarrow 1^-.$$

Differentiating the leading term yields

$$G'_X(t) \sim K(\gamma) [-\ln(1 - t) - 1], \quad t \rightarrow 1^-,$$

and hence  $G'_X(1^-) = \infty$ . Therefore the Discrete Cauchy Distribution does not possess a finite mean, which is fully consistent with the regularly varying tail  $p_x(\gamma) \sim Kx^{-2}$ .

**Corollary 2.4.** *All moments of order greater than or equal to one are infinite for the Discrete Cauchy Distribution (2.2). This property mirrors the continuous Cauchy distribution, in which all positive moments also diverge; however, in the discrete setting this conclusion follows directly from the behavior of the PGF near  $t = 1$ .*

### 3 Conclusion

In this paper, we investigated the PGF of the Discrete Cauchy distribution. An approximate analytical representation of the PGF was derived and expressed in terms of the dilogarithm function. The asymptotic expansion near  $t = 1$  exhibits a logarithmic singularity, which reflects the heavy-tailed structure of the distribution.

Furthermore, the divergence of the derivative of the PGF at  $t = 1$  indicates that the distribution does not possess a finite mean, in agreement with the regularly varying tail behavior of the PMF.

The generating-function approach developed in this work provides a convenient analytical framework for studying discrete heavy-tailed models arising from stable laws. In particular, the asymptotic behavior of  $G_X(t)$  near  $t = 1$  offers a compact criterion for identifying discrete heavy-tailed models with tail index two and may serve as a useful analytical tool in statistical modeling and goodness-of-fit analysis for heavy-tailed count data.

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